

Lecture 3

- Images on web page
- Image file types
 - vector vs bitmap
- Resolution
- Image size
- Scaling image using attributes
- Image maps
- Video and audio
- Browser support for media formats
- Tables

Images on web page

```

```

- `src` attribute
 - Unix-style file path, relative and absolute paths
- `alt` attribute for accessibility
 - semantic information of the image, explaining the content of the image
 - value of alt varies depending of the role of images on a web page
 - informative, decorative, functional, image of text, complex images, groups of images, image maps
- `title` attribute for additional information
 - whereas `alt` is a source of alternate information
- Copyrights not to be forgotten! [Advanced image search with CC license](#)
- [Douglas Crockford on the importance of images](#)



About image file types

- What kind of images can (/should) be used ?

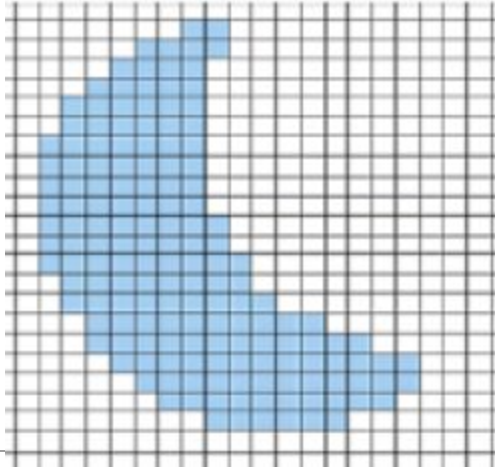
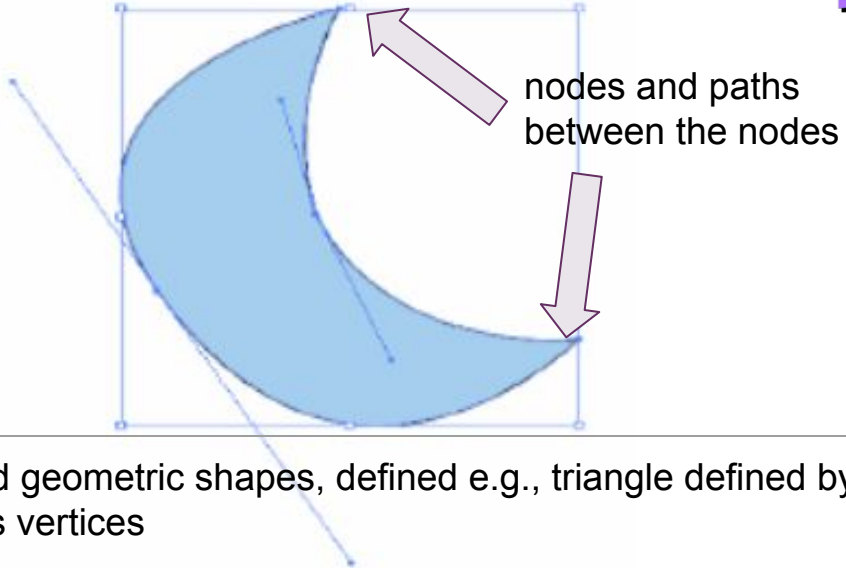
Understanding the Most Popular
Image File Types and Formats

Written by Editorial Team on December 15, 2016

- In short <https://1stwebdesigner.com/image-file-types/>

- **jpg** images compress well, but don't support transparency
- **png** images support transparency, so they are useful for non-rectangular images
- how to transform image transparent ?
 - Paintshop, GIMP, Photoshop,
- **gif** for small animations
- **webp** - in 2010, Google-invented, space-savvy, lossless/lossy image-type, >90% browser coverage(2021)
- **svg** vector graphics=> extremely scalable
 - vector graphic programs (Inkscape, Adobe Illustrator, etc.) can be used to create SVG
 - also by hand, using SVG markup language
 - extra benefit: instructions for laser cutter

Images: raster vs. vector graphics

	
grid of pixels	lines and geometric shapes, defined e.g., triangle defined by giving its vertices
dependent on resolution	independent on resolution
quality and sharpness degrade in scaling	quality remains unaffected in scaling
large file sizes	compact files
formats like .jpg, .png, and .gif	.svg (other, not web compatible vector graphics include .pdf, .ai, and .ps)
pixel-oriented	object-oriented, also mixing HTML elements is possible with foreignObject (check contentModel.svg in demos)

Scalable Vector Graphics

- XML-based W3C standard, developed since 1999
- The latest SVG2 specification is Sep 2018:
 - <https://www.w3.org/TR/SVG2/>
- Powerful graphics capabilities
 - high performance, full animation support
 - controllable with CSS and JS
 - An example: <https://www.w3.org/TR/SVG2/images/conform/smiley.svg>
- Pixel-based versions are costlier than vectors, e.g., a diagonal blue line on 640x480 window
 - As a bitmap, costs ~3000 bytes
 - As a vector, ~20 bytes.
- Bitmaps have problems with resolution and colors when scaled
 - for example, posters in the size of A0 **pixelate** easily

Resolution

- The term resolution is used in two contexts
 - 1) accuracy of the image when printed or displayed
 - number of pixels per inch (ppi), e.g.
 - image printed with 256 ppi produces reasonable quality
 - typical resolutions of displays are 72 – 100 ppi
 - 2) in everyday language: dimensions of image or display
 - number of pixels in width x height
 - for example 8 megapixel camera shoots 3456 x 2304 pixel photos
- Increasing the number of pixels
 - on screen it always increases the size of the image
 - in printed pictures can increase either the accuracy or the size of the image

Resize for speed

- Dimensions of an image
 - measured in pixels
 - for example, the 8 megapixel camera photo
 - when displayed, its physical size depends on the used resolution
 - on paper dimensions ~ 34 x 23 cm
 - on display, dimensions ~100 x 67 cm
 - lots of wasted information to upload to a web page
 - the image can be scaled into desired size on web page
 - both HTML and CSS can be used for that (next dia)
 - still, the image should be resized when used on web (Responsive images)
 - in an image processing program
 - Paintshop, GIMP, Photoshop, ... or Paint (!)
 - Or SVG and Inscape!!!

Scaling the image with attributes

- Layout with CSS, but image size can be defined with the attributes `width` and `height` as well

```

```

- Size of an image is a layout (formatting) feature, so why this is allowed?
 - allowed in order to avoid cluttering the CSS sheet with a different size for each image
 - Actually, there is no need to compute both dimensions
 - one is enough
 - the other automatically computed to maintain the aspect ratio
- ```

```
- however, computing both makes loading the page faster
  - best option is to resize the image into its final size before using it on page



# ... and from CSS

- Resizing of an image should sometimes be done in CSS
  - to prevent adding the same width and height attributes to all images
  - in a situation like this



# Image maps

- We can hide a map into an image
  - defines areas in the image
  - the areas can be attached to a link
- Map: inside a `<map>`-element `<area>`-element
  - each `<area>`-element defines an area in the picture define the areas
  - image and the map are connected using the `usemap` -attribute

```
<map name="aula-map">
 <area shape="circle" coords="717,287,13" href="img/kello.jpg" title="Kello" alt="Clock">
 <area shape="rect" coords="655,200,690,225" title="Right,here" href= "..." alt="Exit sign">
 <area shape="rect" coords="673,274,688,285" title="Exit" href= "..." alt="Exit sign">
 <area shape="rect" coords="678,294,690,303" title="Yess, you found it too! " href="..." alt="Exit sign">
</map>
```

- On-line tools: <https://www.image-map.net/>

# Video (and audio)

`<video src="sourcefile">`

- HTML5 supports natively certain formats
  - no need for plug-ins
- The original idea in HTML5 was
  - to make the video (analogously to audio) embedding as simple as adding an image
  - one format (ogg) for both video and sound

```
<video src="media/video.ogv"> </video>
```

```
<audio src="media/track.ogg"> </audio>
```

# Browser support for video formats

- In the beginning, did not quite work out that way (Apple, Nokia)
  - in order to work in all browsers, videos need to be at least in two formats
    - because of patent issues support not built in all browsers
  - browsers may rely that the support is available from the OS or the hardware
  - videos at least in two formats
- Embedding content from e.g. YouTube and Maps easy
  - services often provide ready to use snippets



Browser  
compatibility

## **Solution 5-2: Detecting video support across browsers**

We have already discussed the wide range of video formats available on the Web these days earlier this chapter.

The first specifications of HTML5 established that browser support for video elements had to be based on two formats: Ogg Vorbis (for audio) and Ogg Theora (for video). This declaration caused quite a stir among the big players, such as Apple and Nokia, so much so that the W3C removed any reference to audio and video formats from the specifications.

This choice obviously led to format chaos on the Web. MPEG4 (H.264), Ogg Theora, AVI, and VP8/WebM are all ready to wage war in the hopes of becoming the standard of the future. Even the browsers have started taking sides. Figure 5-4 summarizes the support provided for video containers in the various browsers:



# Providing several formats

- Several `<source>` (+`<type>`) elements
  - in this case the `src="url"` attribute is left out from the `<video>`-element
  - if present, it overrides the effect of `<source>`-elements

```
<video>
 <source src="media/video.webm" type="video/webm">
 <source src="img/video.mp4" type="video/mp4">
 <p>Sorry, your browser isn't able to show the video.</p>
</video>
```

- browser steps through the `<source>` element list
  - steps out from the `<video>` element when finds a supported format
- Attributes to
  - preload the video, size the area, provide controls, provide a front page image, muting, looping, etc.
- Subtitles can be added to video using `<track>` element

# Sound

- Analogically, sounds are embeddable with <audio>-element
  - playing sound file from the server or from some other address in web

```
<audio autoplay="autoplay" loop="loop">
 <source src="audio/ella.wav" type="audio/wav">
 <source src="audio/ella.mp3" type="audio/mp3">
</audio>
```

```
<audio controls="controls">
 <source src="media/jethro-tull.wav" type="audio/wav">
 <source src="media/jethro-tull.mp3" type="audio/mp3">
</audio>
```

# Tables (simple)

```
<table>[<tr>[<td>cell</td>]*</tr>]*</table>
```

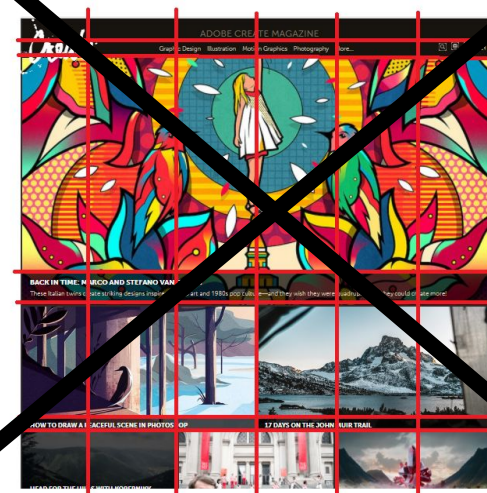
- Used to present information when the nature of data is tabular
  - data with column and/or row headers and data cells
- Not to be used as a layout tool
  - html is about content, not about layout
  - Accessibility! (screen readers get confused)

Rakentaminen

Rakentaminen

[Avaa taulukko suurempana](#)

Valmistuneet rakennukset	Yksikkö	2013	2014	2015	2016
Kaikki rakennukset	milj. m <sup>3</sup>	34,26	33,77	30,89	31,7
Asuinrakennukset	milj. m <sup>3</sup>	12,13	11,64	10,59	10,5
Liike- ja toimistorakennukset	milj. m <sup>3</sup>	4,60	4,80	5,11	5,7
Teollisuus- ja varastorakennukset	milj. m <sup>3</sup>	7,55	6,90	6,70	7,5
Muut rakennukset	milj. m <sup>3</sup>	9,98	10,44	8,49	7,8
Valmistuneet asunnot					
Kaikki asunnot	kpl	29 750	29 295	28 088	30 251
Erilliset pientalot	kpl	9 793	8 365	7 610	7 051
Rivi- ja ketjutalot	kpl	3 751	3 470	3 285	3 571
Asuinkerrostalot	kpl	15 356	17 184	16 972	18 949
Muut kuin asuinrakennukset	kpl	850	276	221	64





# Tables

<code>&lt;table&gt;</code>	defines a table element
<code>&lt;tr&gt;</code>	defines a row element inside a table
<code>&lt;td&gt;</code>	defines a cell element inside a row
<code>&lt;caption&gt;</code>	is used to give a title to the table     - when used, should be used as the first child element of <code>&lt;table&gt;</code>    - can also be helpful for people who have difficulty understanding the content or use assistive technology (as a summary of a table)
<code>&lt;th&gt;</code>	a row or a column header
<code>&lt;col&gt;</code>	can be used if needed to refer to a column
<code>&lt;colgroup&gt;</code>	used to refer groups of columns or     - to clarify the column structure of the table

- In addition to global attributes, `border` attribute is allowed
  - layout done still using CSS

# Tables (more complex)

```
<table>[<caption>Desc</caption>]
[<thead><tr>[<th>cell</th>]*</tr></thead>]
[<tbody><tr>[<td>cell</td>]*</tr>*</tbody>]*
[<tfoot><tr>[<td>cell</td>]*</tr></tfoot>]
</table>
```

- caption for a description
- in addition to colgroups option for row groups with thead, tbody and tfoot
  - thead for column titles, columns marked with <th> elements
    - The text in <th> elements are bold and centered by default.
  - tbody for content (note an option of having multiple tbodies)
  - tfoot for summarizing info

## The problem with `<col>`

The `<col>` element seems like a very handy element for styling columns in

a table — and it does have some benefits. But there is a big problem: the

individual table cells aren't actually contained inside the column. The

`<col>` element only serves to provide information about columns: it isn't the column itself.

[HTML tables](#) are defined by their [rows](#),

not by columns. The result of this is that any style applied to a row will override any style applied to column.

Further complicating things is that [only a handful of CSS properties can be controlled via the `<col>` element](#).

The properties that can be controlled are:

- border
- background
- width
- visibility

So you can control the cell color

(background-color) in a column, you cannot control the color of the text. And if one of your rows is colored, the row color will override the column color.

Read more: <https://html.com/tags/col/#ixzz5htLbzyEE>